

Project Gutenberg language diversity over time

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Abstract

This article analyses the linguistic diversity in Project Gutenberg’s digital library collection over time by analysing books added to the Gutenberg collection over time. The data was pre-processed by downloading the metadata of all Project Gutenberg books from the gutenberg.org website on 06-01-2025. The results indicate that the proportion of English books haven been decreasing over time, while non-English books have maintained a relatively stable addition rate after an initial quick rise in 2004.

1 Introduction

This study analyses the linguistic diversity in Project Gutenberg’s digital library collection over time, examining changes in both English and non-English books added to the Gutenberg Collection per year. It aims to answer what the difference is in the proportion of languages added to Project Gutenberg over time. This research analyses the works added year over year to the Gutenberg collection, showing the changes in addition rates of differing languages. The hypothesis is that over time, the proportion of English books will decrease compared to non-English books, reflecting an increase in internet availability and interest

in preservation of historical books and a decrease in available public domain books written in English.

2 Related Work

Several studies have examined linguistic diversity in digital libraries and Project Gutenberg specifically:

1. Whitt (2018) explored multilingualism on the web and Project Gutenberg’s early efforts at linguistic diversification, providing important historical context for understanding the evolution of digital preservation efforts.
2. Gerlach and Font-Clos (2020) analysed a curated Project Gutenberg database, revealing a strong bias toward Western languages, with English comprising 81% of all books. Their work established baseline metrics for understanding linguistic representation in digital libraries.
3. Budzise-Weaver, Chen, and Mitchell (2012) compares four multilingual digital libraries, including Project Gutenberg, using a digital library evaluation framework. They noted that the libraries all used volunteers to translate books and manage other parts of the library, with Project Gutenberg being one of the most

active on this front, having more languages than the other libraries.

Pallot, Martínez-Carreras, and Prinz (2010) discusses a model for collaborative work, and the factors that influence the efficiency of the collaborative work. Several of these factors could have influenced the additions of books to Project Gutenberg. For example, the Organizational and Spatial structural distance types or the Temporal, Relational and Lingual social distance types. These factors could have influenced the addition of books to Project Gutenberg, as the volunteers that add books to Project Gutenberg are not directly managed by Project Gutenberg. This could make communication between Project Gutenberg and the volunteers difficult.

Project Gutenberg also knows about the disparity in languages in its collection. For example, on the frequently asked questions pages on the gutenberg.org website they explain how books are added, namely by volunteers. This makes it difficult for Project Gutenberg to directly change which books are added, as they can't dictate what these volunteers add.

3 Data

Pre-processing The data pre-processing was done by downloading the metadata of all Project Gutenberg books from the gutenberg.org website on 06-01-2025. This data can be downloaded in the form of a CSV file, which contains the relevant fields for this article: the date added and the language. Books that included multiple languages, either due to being a translation that includes the original text or being a translation dictionary, are counted for both languages. The python script used for processing the data will be available in the Github repository linked in the Appendix. Due to not being completed, the year 2025 was excluded. As only the language codes were given with the metadata,

Decade	English	Other
1970-1980	9	0
1881-1990	1	0
1991-2000	1861	79
2001-2010	27375	5526
2011-2020	22767	6703
2021-2024	7970	2759

Table 1: Books added to Project Gutenberg per decade

language codes were manually linked to their respective language. This resulted in a matrix for book additions per year for every language, from the years 1971 to 2024, with 70 languages. The non-English books were all summed, creating the distribution graph shown in figure 1. Books added per decade are shown in table 1. The complete data matrix is shown in table 2 in the Appendix, with the total books per language shown in figure 2, also in the appendix.

4 Predicted Results

As seen in figure 1, the proportion of English to non-English books has significantly changed over time, particularly after the large increase in books added in 2004. But while English books experienced a sharp decline after this peak of over 6,000 additions, non-English books maintained a relatively stable addition rate of 600–800 books per year following 2004. Furthermore, English book additions have begun to steadily decline after the 2004, if inconsistently. If this pattern continues, the gap between English and non-English book addition rates will likely become narrower in the future.

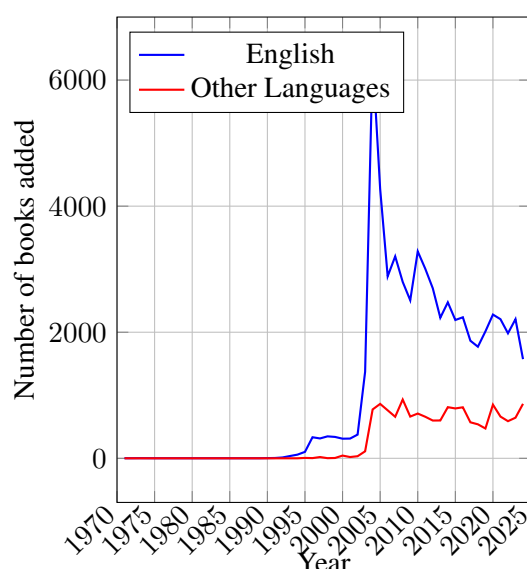


Figure 1: Number of books added to Project Gutenberg per year for English and non-English languages

Discussion The trend shown in the data strongly supports the hypothesis that the proportion of English books would decrease compared to non-English books. The decline in English book additions could be attributed to the fact that most English public domain books have already been entered, with remaining books becoming increasingly rare. The stability in non-English additions suggests growing international attention for digital preservation efforts. This could be due to several factors: increased global internet accessibility, greater awareness of Project Gutenberg internationally, and improved or more accessible optical character recognition (OCR) technology for non-Latin scripts, making it easier to participate. Additionally, as many English books have already been digitized, volunteers might be increasingly focussing on preserving texts in other languages.

5 Conclusion

This article has demonstrated a shift in the linguistic composition of Project Gutenberg’s digital library collection over time. It reveals that while English-language content has been the majority of additions to the Gutenberg collection, there has been a slow but consistent decline in the rate of English book additions since 2004. On the other hand, non-English books have maintained relatively stable addition rates per year since 2004. The findings support the initial hypothesis that the proportion of English books would decrease relative to non-English books over time. This trend can be attributed to several factors, including the running out of available English public domain books, increased global participation in digital preservation efforts, and technological advancements in handling non-Latin scripts. These results have important implications for the future of digital libraries and preservation efforts. As Project Gutenberg continues to evolve, the data suggests a natural progression toward a more linguistically diverse collection, better representing the global literary heritage. Future research could explore the specific causes of the decline in English book additions, the reason for the peak in 2004, and the factors driving the consistent addition of non-English books.

References

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6 Appendix

Data

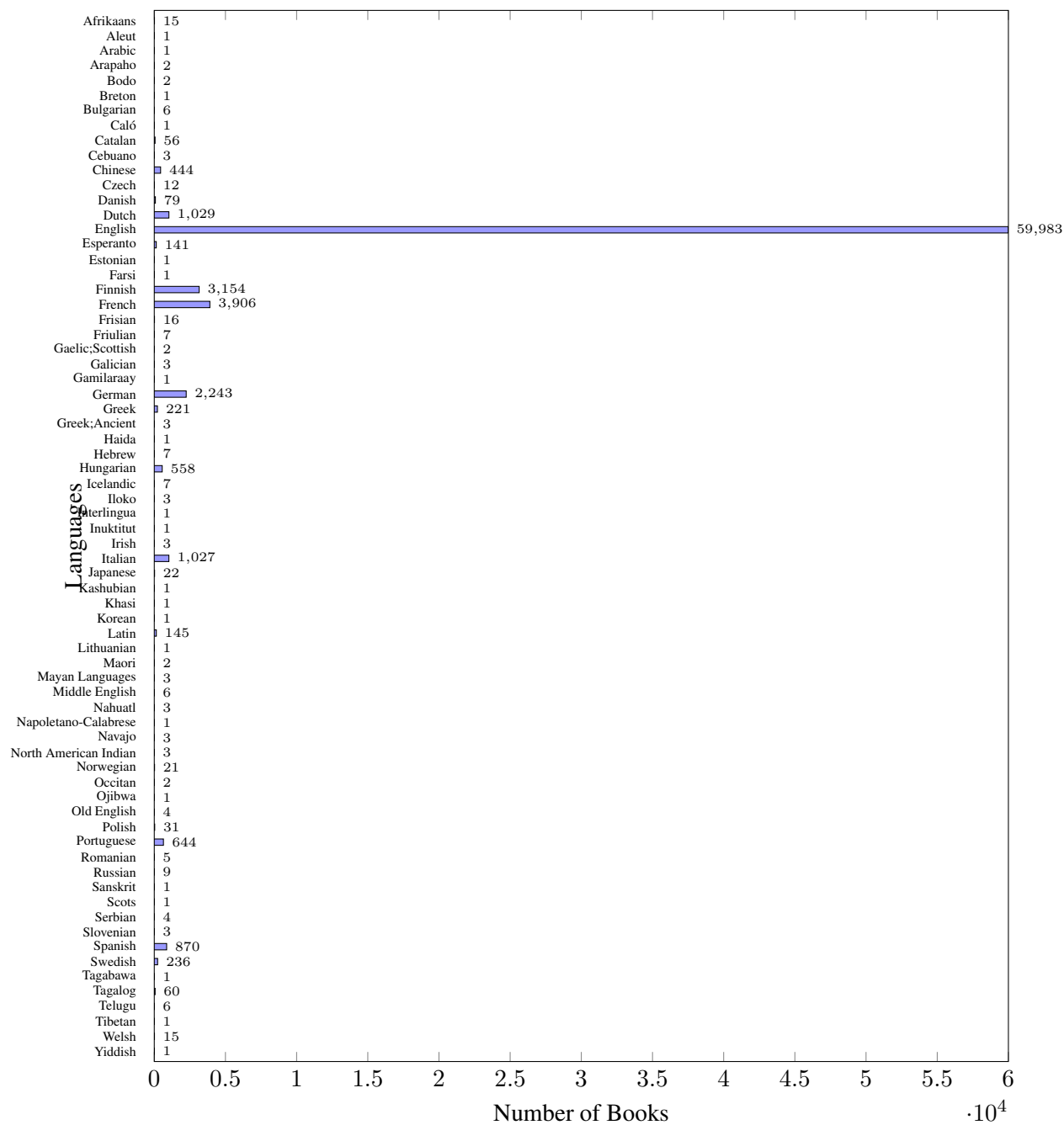


Figure 2: Total books per language in the Gutenberg Corpus

Language	1971	1972	1973	1975	1976	1977	1978	1979	1988	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024		
English	1	1	2	0	0	1	1	1	1	1	5	13	36	58	102	334	315	348	339	311	313	376	1373	6563	4273	2847	3203	2798	2307	3282	3007	2697	2230	2474	2194	2237	1865	1770	1865	2207	1983	2207	1573		
French	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	2	2	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	201	160	133	168	210	
Finnish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	114	142	91	65	64	37	41	82	61	73	206	267	326	230	154	150	210	129	86	229	393		
German	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	37	5	28	26	116	139	61	85	59	67	104	91	111	79	181	172	159	117	74	110	115	83	97	67	58		
Dutch	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	6	80	71	87	62	70	76	54	55	40	40	36	40	25	27	25	8	32	55	65	42	30		
Italian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	7	15	12	24	14	32	40	55	39	100	116	87	45	40	41	58	60	37	24	36	45			
Portuguese	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	25	44	27	42	52	29	22	35	43	57	46	44	35	45	47	43	27	28	38	43			
Provençal	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
Hungarian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	6	28	34	100	118	92	120	20	9	4	5	1	5	4	3	3	35	1	32	17	5		
Chinese	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	2	7	6	0	46	32	10	1	0	18	53	0	96	119	92	23	45	
Swedish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	10	3	55	0	78	247	11	0	0	2	3	0	6	38	31	23	19	12	9	11	1	0	1	0	0
Greek	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	9	14	9	5	11	11	7	0	5	0	6	38	31	23	19	12	9	11	1	6	7	
Latin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
Esperanto	0	0	0	0	0	0	0	0	0	0	0	0	0	5	1	4	0	0	0	0	0	3	1	17	9	11	10	5	9	5	7	4	5	5	12	2	7	5	2	2	5	7	0	0	
Yiddish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
Tamil	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Tagalog	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Catalan	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Polish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Japanese	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Norwegian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Frisian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Irish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Welsh	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Malay	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Czech	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
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Ukrainian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Polish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Arabic	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Hebrew	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Indonesian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Malay	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Tagalog	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Yiddish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Ukrainian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
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Arabic	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Hebrew	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Indonesian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Malay	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Tagalog	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
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Ukrainian	0	0	0	0	0																																								

Github

Repository

[https://github.com/DirkvanW/Introduction-
Research-Methods](https://github.com/DirkvanW/Introduction-Research-Methods)